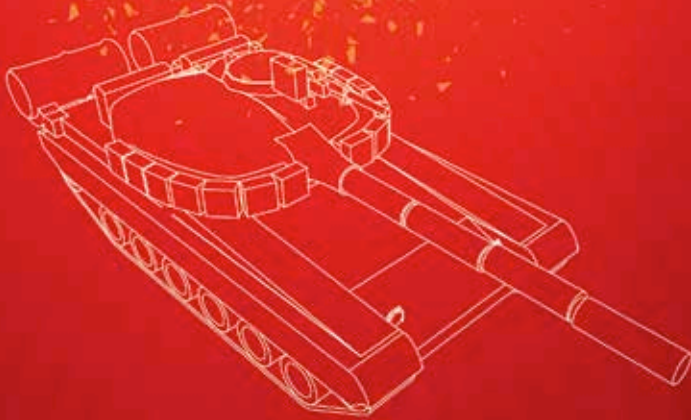


LAND

DOING THE DIRTY WORK



The civilian and commercial applications of unmanned ground vehicles (UGVs) or autonomous land vehicles (ALVs) are staggeringly wide. UGVs are used to harvest crops, transport heavy materials for manufacturing, handle warehousing tasks, map out mining tunnels and aid emergency workers in search and rescue operations. The Nasa rovers on Mars are highly autonomous UGVs. Modern armies have a number of tracked and wheeled UGVs in use for mine clearance, explosive ordnance disposal, clearing rubble or bulldozing, and evacuating wounded soldiers from combat zones to free up two soldiers holding a stretcher. There are also UGVs for moving supplies and gathering intelligence from hazardous areas. These are all in a variety of form factors, from the four legged BigDog that can follow soldiers through hazardous terrain loaded with supplies, to tiny threaded vehicles that look like miniature tanks, to other UGVs that look more like the conventional trucks,

armoured vehicles and tanks. These UGVs in deployment are all meant to do the dirty work that is considered too hazardous for humans. The next generation of UGVs in development and testing though have advanced capabilities including crowd control, convoy protection, perimeter defence, border patrol, surveillance and evacuation.

The military applications of UGVs have a long history though. In the first world war, a number of inventors came up with tethered vehicles for deploying explosives, including the French Land Torpedo Crocodile Schneider. In 1942, the German Military developed the Goliath tracked mines that were tethered, single use petrol or electric powered vehicles meant to deliver explosives to targets. Neither of these designs were considered particularly effective, especially in comparison to the tanks that saw their first use in World War I.

Since the 1960s, DARPA began developing UGVs in the US, with a number of early designs allowing for remote steering, with computers on board for rudimentary forms of obstacle avoidance. The birth of the

modern UGV can be considered to have taken place in May 1985. This was a vehicle that looked something like an ice cream truck, with a rooftop camera on board. Within the truck were six racks of computers that analysed the images from the rooftop camera to autonomously steer the UGV down the narrow dirt road near the Rocky Mountains. This ALV became the platform on which more modern UGVs were developed. With the march of miniaturisation, it became possible to reduce the footprint of the computers on board, in terms of both space and energy use. Miniaturisation also allowed for progressively adding some autonomous capabilities to existing land vehicles. The initiative was known as Convoy Active Safety Technology (CAST) and allowed a driver under fire in a convoy to lock the vehicle onto others in the convoy, and take the necessary actions to counter the enemy combatants. This locking on concept was later on used in the Squad Mission Support System (SMSS), a land vehicle that followed soldiers wherever they went, while being able to carry cargo, wounded soldiers or energy supplies. This six wheeled SMSS made its combat debut in Afghanistan in 2011, where it quickly earned the moniker of the "Ox" for being so reliable. The early ALV, CAST and SMSS were all developed by the company that is now Lockheed Martin along with DARPA.

BAE Systems, another major defense contractor has developed



The BigDog loaded with supplies

Credit: Boston Dynamics